Questions: Introduction to logic (maths)

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Summary

A selection of questions for the study guide introducing propositional logic (maths).

Before attempting these questions, it is highly recommended that you read Guide: Introduction to logic (maths).

Q1 Propositions

Which of the following statements are propositions?

1.1. "$3 < 5$"

1.2. "$5 < 3$"

1.3. "$x < 5$"

1.4. "5 is even."

1.5. "y is odd."

1.6. "$2 \cdot 3 = 6$"

1.7. "$4 \cdot 3 = 10$"

1.8. "$2 \cdot 3 < 5x$"

Q2 Connectives

What is the logical structure for the following statements?

2.1. "If $x < 3$, then y is even."

2.2. "x is odd and positive."

2.3. "z is not -1."

2.4. "y is even or odd."

2.5. "z is positive if and only if y is negative."

2.6. "If $x < 0$, then y and z are even."

Q3 Truth tables

Give the truth tables for the following logical statements:

3.1. $P \to Q$

3.2. $\neg P$

3.3. $(P \vee Q) \wedge \neg Q$

3.4. $\neg(P \leftrightarrow Q)$

3.5. $\neg(P \wedge Q)$

3.6. $(P \to Q) \to R$

3.7. $\neg(P \vee (Q \vee R))$

3.8. $(P \leftrightarrow Q) \vee (\neg R)$

Q4 Equivalence

Using truth tables or otherwise, are the following pairs of statements logically equivalent?

4.1. $\neg(P \wedge Q)$ and $(\neg P) \vee (\neg Q)$

4.2. $P \to Q$ and $Q \to P$

4.3. $P \to Q$ and $(\neg P) \vee Q$

4.4. $P \to Q$ and $\neg(P \vee Q)$

After attempting the questions above, please click this link to find the answers.

Version history and licensing

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